

Parallel Computing with MPI

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Today's goals

Complete any 3 learning items - 0/3

Complete a practice item

Complete 2 readings - 0/2

Module 1

Introduction to Parallel Computing and MPI

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Welcome to the Course

Course Updates and Accessibility Support

Reading • 1 min

Earn Academic Credit for your Work!

Reading • 10 min

Course Support

Reading • 10 min

Course Overview

Video • 2 min

Foundations of Parallel Computing

Introduction to Parallel Computing

Video • 6 min

Thinking in Parallel

Video • 5 min

Message Passing and Synchronization

Introduction to Message Passing

Video • 4 min

Synchronization

Video • 3 min

Module Assignments

Module Quiz

Graded Assignment • 15 min

Distributed Weather Prediction System Using MPI

Programming Assignment • 1h

Module 2

Advanced Communication Techniques in MPI

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Module 3

Performance Optimization in Parallel Computing

▼

Module 4

Advanced MPI Concepts – Communicators and Derived Datatypes

▼

Module 5

Parallel I/O in MPI and HDF5 for High-Performance Computing

▼

Earn Academic Credit for your Work!

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0:13 / 4:29

2x

You can earn academic credit for this course through the University of Colorado at Boulder as part of the [Master of Science in Data Science \(MS-DS\)](#) [🔗](#) program.

CU Degrees on Coursera

With performance-based admissions and no application process, these programs are ideal for individuals with a broad range of undergraduate education and/or professional experience in computer science, information science, mathematics, and statistics.

CU degrees on Coursera offer stackable graduate-level courses, graduate certificates, and fully accredited master's degrees. You will earn the same credentials as our on-campus students while enjoying targeted courses, short 8-week sessions, and pay-as-you-go tuition. Our innovative performance-based admissions allow you to earn program admission simply by proving you can do the work. We never ask for transcripts, letters of recommendation, GRE results, or TOEFL scores. Successfully complete a special series of for-credit courses called a "pathway specialization" to gain program admission while earning credits that apply directly toward your degree.

Learn more about [flexible online degree programs from the University of Colorado Boulder](#) [🔗](#).

Master of Science in Data Science

The MS-DS is an interdisciplinary degree that brings together faculty from CU Boulder's departments of Applied Mathematics, Computer Science, Information Science, and others. [Learn more about the Master of Science in Data Science](#) [🔗](#) program.

Upgrading to For-Credit

You can upgrade to the for-credit version of this course at any point up until the enrollment deadline. Please refer to the [CU Degrees on Coursera Calendar](#) [🔗](#) for session schedules. Work that you complete before upgrading will transfer when you enroll in the for-credit course.

What's the Difference Between the Non-Credit and For-Credit Version of this Course?

Coursework

The course content and assignments remain the same for both the non-credit and for-credit options, making upgrading a breeze. Your progress in the non-credit course will count towards the for-credit version, allowing for a smooth transition. Typically, the only difference in course content you will find after upgrading is an additional module containing a final exam and/or project. This allows you to assess your ability to pass the course before committing to tuition fees.

Technical and Learning Support

The level of support provided by course facilitators is the primary distinction between the non-credit and for-credit experience. If your goal is to eventually upgrade to the for-credit version, it's worth noting that enrolling sooner rather than later provides certain benefits. This includes additional support and feedback only available in the for-credit version.

Non-Credit Learners

Similar to other courses on Coursera, the non-credit version is a self-paced and asynchronous learning experience. You can seek out support from peers in the discussion forums for questions about course assignments or materials, explore the Coursera [Learner Help Center](#) [🔗](#) or chat or email with Coursera's [support team](#) [🔗](#) for platform questions or concerns. For non-urgent concerns, you can [flag](#) [🔗](#) items in the course, for instance readings, quizzes, etc. Facilitators routinely review feedback and flags throughout the course. Additionally, course facilitators monitor the [Course Support Forum](#) [🔗](#) and will address issues that impede learner progress. Please be aware that while enrolled in the non-credit version, course facilitators and instructors cannot view your grades or offer feedback on your assignments.

For-Credit Learners

For-credit students are invited to attend weekly live virtual office hours with their course facilitator(s). This provides an excellent opportunity to ask questions about course content and receive feedback on your work. Facilitators also respond to questions and comments in all for-credit discussion forums. They have access to your grades and individual assignments, enabling them to provide support with grading and technical concerns.

Additional For-Credit Benefits

As a for-credit learner, you will have access to a dedicated email alias offering degree planning and technical support from program staff. Your CU tuition also grants you access to additional CU resources, including online library and alumni/career services, [CU on Coursera](#) [🔗](#) membership, and a CU student email address.

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